Automatic GFL/GFM Mode Switching

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Introduction

Grid-following (GFL)

- PLL-synchronized current control
- Requires an external voltage-angle reference
- Effective while a strong grid reference exists

Grid-forming (GFM)

- Establishes voltage angle and frequency
- Supports weak-grid and islanded operation
- Can replace the lost reference after SG disconnection

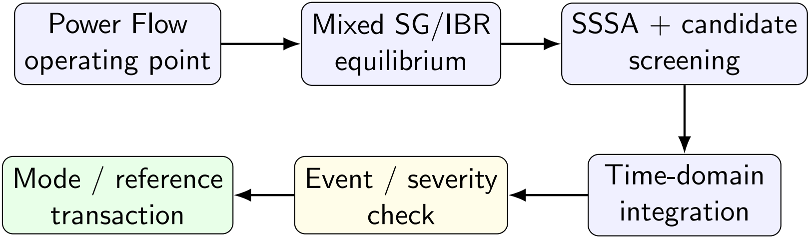
Project Objectives

1. Develop one in-house analysis chain for **Power Flow**, **SSSA**, and **Time-Domain Simulation**.
2. Model four converter resources with a common plant and selectable **GFL/GFM** controller branches.
3. Determine admissible GFM configurations from the network condition instead of fixing the number of forming units in advance.
4. Verify automatic mode transitions, island support, and SG reconnection under one disturbance chronology.

Scope

IEEE 14-bus test system, one synchronous generator, four dual-mode IBRs, automatic reference support, and event-driven nonlinear simulation.

Overall Analysis and Switching Workflow



Runtime decision

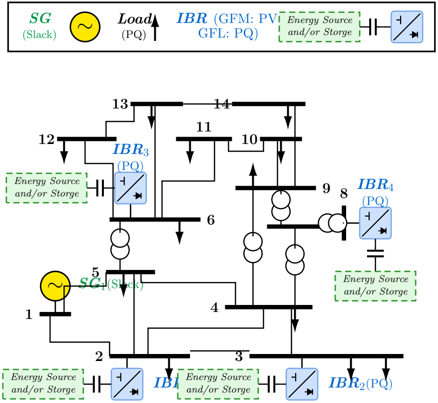
No trigger: advance to the next TS step. Triggered event: certify and commit the mode/reference transaction only after current continuity and right-limit KCL convergence.

Case Study: IEEE 14-Bus with 1 SG + 4 Dual-Mode IBRs

Resource mapping

Resource	Bus
SG1	1
IBR1	2
IBR2	3
IBR3	6
IBR4	8

- SG1: normal synchronous reference
- IBR1–IBR4: dual-mode GFL/GFM converters



Power Flow Formulation

Following the polar form used in Hadi Saadat,

$$Y_{ik} = |Y_{ik}|\angle\theta_{ik}, \qquad V_i = |V_i|\angle\delta_i$$

$$P_i = \sum_{k=1}^{n} |V_i||V_k||Y_{ik}|\cos(\theta_{ik} + \delta_k - \delta_i),$$
$$Q_i = -\sum_{k=1}^{n} |V_i||V_k||Y_{ik}|\sin(\theta_{ik} + \delta_k - \delta_i).$$

Bus constraints

Type	Specified	Solved
Slack	$\lvert V\rvert, \delta$	P, Q
PV	$P, \lvert V\rvert$	Q, δ
PQ	P, Q	$\lvert V\rvert, \delta$

Newton–Raphson correction

$$\Delta P_i = P_i^{sch} - P_i, \qquad \Delta Q_i = Q_i^{sch} - Q_i$$

$$\boxed{\begin{bmatrix} \Delta \boldsymbol{P} \\ \Delta \boldsymbol{Q} \end{bmatrix} = \boldsymbol{J}_{PF} \begin{bmatrix} \Delta \boldsymbol{\delta} \\ \Delta |\boldsymbol{V}| \end{bmatrix}}$$

$$\boldsymbol{\delta}^{(r+1)} = \boldsymbol{\delta}^{(r)} + \Delta\boldsymbol{\delta}^{(r)}, \qquad |\boldsymbol{V}|^{(r+1)} = |\boldsymbol{V}|^{(r)} + \Delta|\boldsymbol{V}|^{(r)}$$

Small-Signal Stability Analysis (SSSA)

Linear small-signal model

$$\Delta\dot{\boldsymbol{x}} = \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u}$$

For the autonomous stability study used here, no AVR/PSS perturbation input is applied:

$$\boxed{\Delta\dot{\boldsymbol{x}} = \boldsymbol{A}\Delta\boldsymbol{x}}$$

The state matrix is obtained from the linearized DAE:

$$\boxed{\boldsymbol{A} = \boldsymbol{f}_x - \boldsymbol{f}_y\left(\boldsymbol{g}_y \backslash \boldsymbol{g}_x\right)}$$

Modal analysis

$$\boxed{\boldsymbol{A}\boldsymbol{\phi}_i = \lambda_i \boldsymbol{\phi}_i}$$

$$\lambda_i = \sigma_i + j\omega_i, \qquad f_i = \frac{|\omega_i|}{2\pi}$$

$$\zeta_i = -\frac{\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}$$

Stability criterion

$\mathrm{Re}(\lambda_i) < 0$ for every physical mode, together with the declared damping-ratio floor.

Time-Domain Simulation (TS)

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}), \qquad \boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y})$$

Implicit trapezoidal step

$$\boldsymbol{R}_x = \boldsymbol{x}_{n+1} - \boldsymbol{x}_n - \frac{h}{2}\left(\boldsymbol{f}_n + \boldsymbol{f}_{n+1}\right) = \boldsymbol{0},$$

$$\boldsymbol{R}_y = \boldsymbol{g}(\boldsymbol{x}_{n+1}, \boldsymbol{y}_{n+1}) = \boldsymbol{0},$$

$$\boldsymbol{J}_R \Delta \boldsymbol{z} = -\boldsymbol{R}, \qquad \boldsymbol{z} = \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix}.$$

Event handling

- Land exactly on each event time.
- Preserve the left-limit state.
- Apply mode/topology changes atomically.
- Require right-limit KCL convergence.
- Roll back a rejected transaction.

Dual-Mode IBR Dynamic Model

Fixed 17-coordinate switching superset

$$\boldsymbol{x}_{IBR} = \begin{bmatrix} \boldsymbol{x}_p^T & \boldsymbol{x}_{GFL}^T & \boldsymbol{x}_{GFM}^T & I_{dc} \end{bmatrix}^T$$

$$\begin{aligned} \boldsymbol{x}_p &= [i_d \; i_q \; V_{dc}]^T, \\ \boldsymbol{x}_{GFL} &= [\delta_{PLL} \; \xi_{PLL} \; \xi_P \; \xi_Q \; \xi_{Id} \; \xi_{Iq}]^T, \\ \boldsymbol{x}_{GFM} &= [\delta_{VSG} \; \omega_{VSG} \; E \; \xi_{Vd} \; \xi_{Vq} \\ &\qquad \xi_{Id}^{GFM} \; \xi_{Iq}^{GFM}]^T. \end{aligned}$$

$$\mathcal{A}_{GFL} = \{1{:}9, 17\}, \qquad \mathcal{A}_{GFM} = \{1{:}3, 10{:}17$$

Inactive controller coordinates are held during each mode.

GFL: PLL-synchronized

$$\begin{aligned} \dot{\delta}_{PLL} &= k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \\ e_P &= \kappa P^\star - P, \\ i_d^\star &= k_{pP} e_P + k_{iP} \xi_P. \end{aligned}$$

GFM: voltage-forming VSG

$$\begin{aligned} M\dot{\omega} &= \kappa P^\star - P - D_v(\omega - 1), \\ \dot{\delta}_{VSG} &= \omega_b(\omega - 1), \\ \tau_E \dot{E} &= k_Q(\kappa Q^\star - Q) \\ &\quad - k_E(E - E^\star). \end{aligned}$$

Automatic GFL/GFM Switching Law

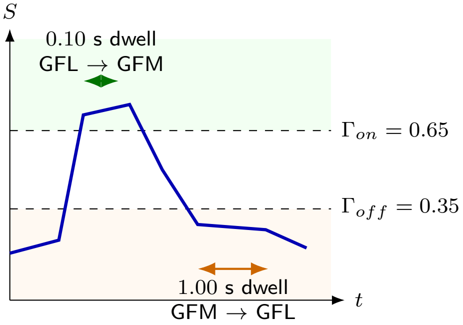
Severity index

$$J_V = \frac{\big||V_i| - V_i^\star\big|}{0.10\,\mathrm{p.u.}},$$
$$J_f = \frac{|f_{COI} - 60|}{0.50\,\mathrm{Hz}},$$
$$S = \mathrm{sat}_{[0,1]}(0.5J_V + 0.5J_f)\,.$$

Hysteretic command

$$q^+ = \begin{cases} \mathrm{GFM}, & S \geq 0.65 \text{ for } 0.10\,\mathrm{s}, \\ \mathrm{GFL}, & S < 0.35 \text{ for } 1.00\,\mathrm{s}, \\ q, & \text{otherwise.} \end{cases}$$

$$\left\|I^+ - I^-\right\|_\infty \leq 10^{-10}\,\mathrm{p.u.}$$



Transition is committed only after

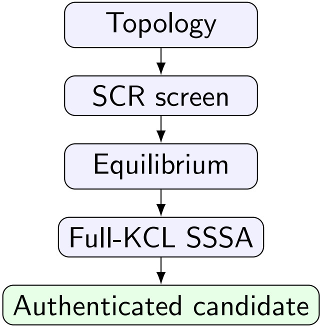
equilibrium/SSSA certification and a converged right-limit KCL solve.

Result 1: GFM Candidate Screening

GFM count	Candidate	Outcome
1	any single IBR	admissible
2	IBR1 + IBR3	admissible
2	**IBR2 + IBR3**	**best margin**
2	other pairs	rejected
3	any triple	rejected
4	all four IBRs	admissible

Observed property

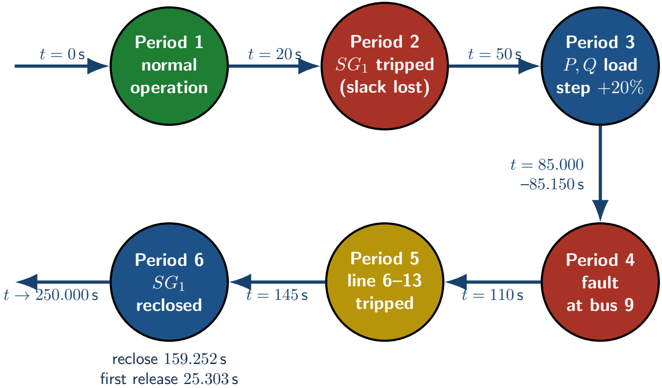
Admissibility is **non-monotonic** in the number of grid-forming converters.



The GFM set is computed from

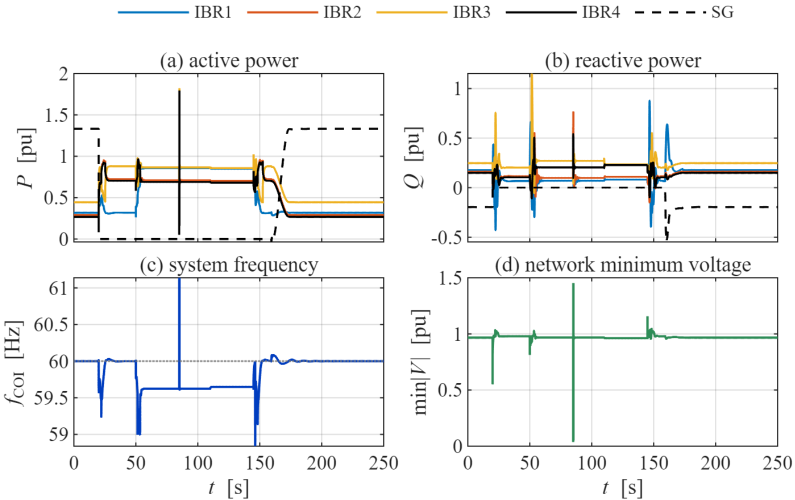
the current network condition; converter count alone does not determine admissibility.

Result 2: Disturbance Scenario



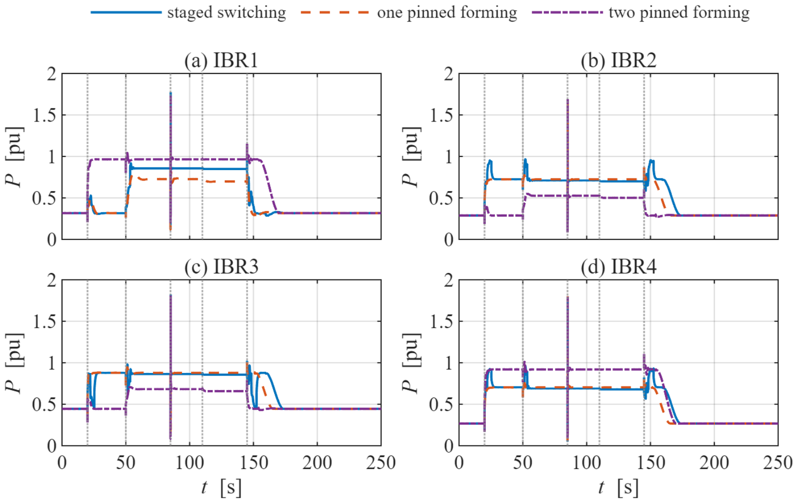
Green: undisturbed; red: source loss or fault; yellow: topology change; blue: restorative action. Scheduled times are case-defined; reclose and release times are read from the accepted event log.

Result 3: Electrical Response



System active/reactive power, COI frequency, and minimum bus voltage over the complete 250 s chronology.

Result 4: Converter Response under Different Policies



IBR1–IBR4 active-power response: staged switching (solid), one pinned GFM (dashed), and two pinned GFMs (dash-dot).

Conclusion and Key Findings

- PF, SSSA, and TS are linked through one nonlinear SG/IBR network model.
- After SG1 is disconnected, at least one voltage-forming IBR is required for the islanded trajectory to continue.
- GFM admissibility is configuration-dependent and non-monotonic in converter count.
- Staged switching reaches the full 250 s chronology and returns all IBRs to GFL operation after SG recovery.

Comparative outcome

- **Locked GFL:** stops at 20 s; no forming source.
- **1–2 pinned GFMs:** reach 250 s; reclose succeeds.
- **4 pinned GFMs:** nonlinear failure at 25.485 s.
- **Staged switching:** 250 s with full hand-back.

Scope of claim: the staged policy is not claimed to minimize every transient peak; its benefit here is adaptive reference support across the complete disturbance chronology.